

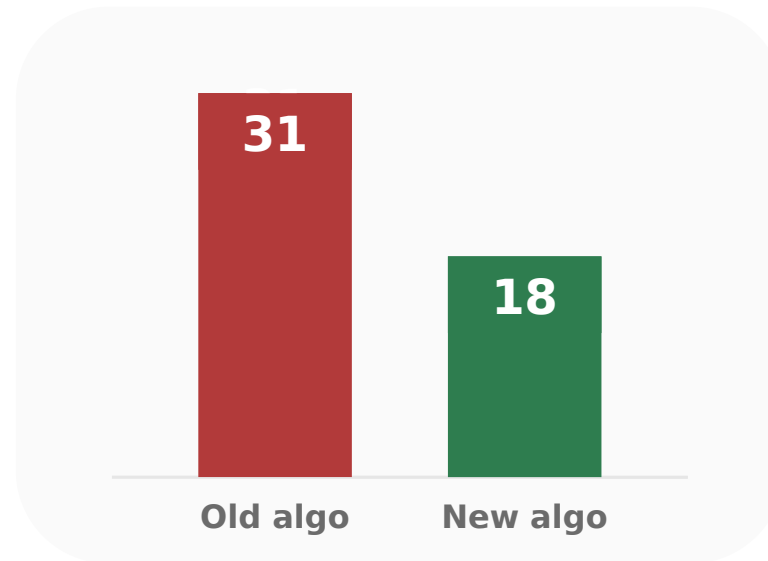
The **Latent** Data World

*Evaluating VLM agents as **visual data investigators***

“Charts are **clues. The task is **investigation**.”**

A 38% speed-up — or a mirage?

SwiftEats rolls out a new AI dispatcher. The headline reads:



Average delivery time (min)

Same week, same dashboard: refunds, cancellations, complaints — all up.

Is the 18 minutes real, or a dashboard mirage?

Four suspects. Which one really explains **31 → 18**?

The agent will distinguish them through visual investigation. Track their state across four beats.

h₁

“Genuine speedup”

The new algorithm really does shorten routes — every order, every zone, every time.

h₂

“Zone selection”

The platform silently abandons distant / low-density zones — only easy orders remain.

h₃

“Denominator wash”

“18 min” counts only successful deliveries — cancellations and no-driver orders vanish.

h₄

“Metric boundary”

“18 min” covers only pickup → drop; click → pickup waiting is moved outside the boundary.

The chart is a **shadow**.

Every dashboard is a lossy projection of a hidden master table.

ABOVE THE WATER · WHAT THE AGENT SEES

Two bars. One number. *Airtight, at first glance.*

missing · **acceptance per zone** *missing* · **order outcome**
missing · **window boundary** *missing* · **failure funnel**

BELOW THE WATER · HIDDEN MASTER TABLE

A DAG of typed columns.
Atomic rows. *Cross-chart arithmetic by construction.*

order_id zone_id algorithm outcome pickup→drop
click→door

The truth has **structure**. The chart reveals only a sliver of it.

Benchmarks test **reading**, not investigation.

Models must answer from one given chart. They cannot decide to look again.

–30

pts on real reasoning

ChartMuseum 2025 — human 93% vs top VLM 63%.

2–3

charts ⇒ collapse

InterChart 2025 — accuracy falls across multi-chart splits.

×

ungrounded answers

ChartPoint 2025 — models fail to point to correct elements.

OLD WAY — PASSIVE SINGLE-SHOT

[chart] → answer

no follow-up · no grounding · “reading the picture”

OUR WAY — ACTIVE, BUDGETED, GROUNDED

[D₀] ∪ request more views → verdict

bbox = receipt · “investigating the world”

To investigate, the agent needs **both halves**.

A hidden ground truth to probe — and the visual moves to probe it with.

THE GAP TO FILL

An investigation needs a **hidden Master Table** — and a **protocol** that queries it through charts.

C1

Builds the **world**

Synthetic data generation

LLM writes Python over a typed SDK → atomic-grain Master Table → 10-30+ chart QA tasks via deterministic SQL projection.

C2

Investigates the **world**

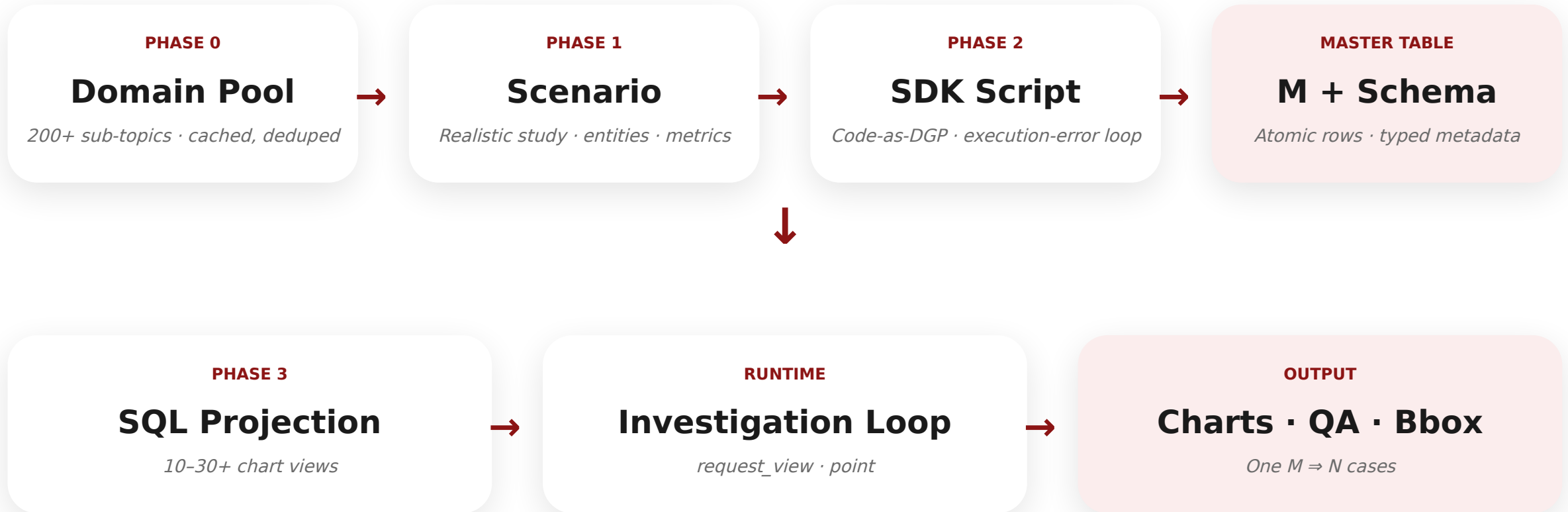
Active chart reasoning

Budgeted visual actions, bbox-as-receipt grounding, four investigation moves: Read · Reveal · Reconcile · Debunk.

C1 builds the world. C2 lets the agent investigate it.

Phases 0-2 use the LLM. Phase 3 is **deterministic**.

No LLM touches the chart or the questions. The Master Table is the contract.



PART · C1

How a Master Table is **written by code.**

Phases 0 → 1 → 2 → 3, walked through with concrete examples.

200+ **fine-grained domains**, cached once.

Embedding-deduplicated and complexity-balanced — the seed for every downstream scenario.

EXAMPLE · SIMPLE

```
{  
  "name": "Coffee shop POS",  
  "topic": "Retail & Services",  
  "tier": "simple",  
  "entities": ["drink", "hour"],  
  "metrics": ["revenue"],  
  "grain": "hourly"  
}
```

EXAMPLE · MEDIUM

```
{  
  "name": "Urban food delivery",  
  "topic": "Logistics",  
  "tier": "medium",  
  "entities": ["order", "zone"],  
  "metrics": ["delivery_min"],  
  "grain": "per order"  
}
```

EXAMPLE · COMPLEX

```
{  
  "name": "Urban rail scheduling",  
  "topic": "Transportation",  
  "tier": "complex",  
  "entities": ["line", "station"],  
  "metrics": ["ridership", "on_time"],  
  "grain": "daily"  
}
```

Generated once. Cached forever. *Stratified sampling preserves complexity balance.*

Domain → one **realistic scenario**.

Entities, metrics, time grain, target_rows. No chart type bound yet.

INPUT · DOMAIN

```
{
  "name": "Urban rail scheduling",
  "tier": "complex",
  "entity_hint": ["line", "station"],
  "metric_hint": [
    {"name": "ridership",
     "unit": "10k pax"},
    {"name": "on_time_rate",
     "unit": "%"}
  ]
}
```

OUTPUT · SCENARIO (LLM)

```
{
  "title": "2024 H1 Shanghai Metro Ridership Log",
  "context": "Shanghai Transport Commission collected
             daily ridership and operational data...",
  "entities": [
    "Line 1 (Xinzhuang-Fujin Rd)",
    "Line 2 (Pudong Airport-Xujing)", ...
  ],
  "metrics": [
    {"name": "daily_ridership", "range": [5, 120]},
    {"name": "on_time_rate", "range": [85.0, 99.9]}
  ],
  "target_rows": 900
}
```

Data is born from **business context**, not visualization templates.

The LLM writes **Python**, not JSON.

Code-as-DGP. Two ordered steps. Eight call types. Each column carries its own DGP.

STEP 1 · COLUMN DECLARATIONS

add_category

categorical · group · parent

add_temporal

time grain · derived calendar

add_measure

stochastic root · $y \sim \text{Dist}(\theta)$

add_measure_structural

DAG-derived · closed-form

STEP 2 · RELATIONSHIPS & PATTERNS

declare_orthogonal

independence across groups

add_group_dependency

root-only DAG cross-group dep

inject_pattern

trend · outlier · ranking reversal

set_realism

noise · missingness · censoring

Closed-form measures · one declaration each · never patched across calls.

SwiftEats — **zone × algorithm** drives both.

Step 1: columns

```
sim.add_category("zone_tier", values=["downtown","mid","peripheral"], group="geo")
sim.add_category("algorithm", values=["old","new"], group="system")
sim.add_measure("distance_km", family="lognormal",
    param_model={"mu": 1.5, "sigma": 0.6})
```

Step 2: zone × algorithm drives accept; both drive p→d

```
sim.add_group_dependency("accepted", on=["zone_tier","algorithm"],
    conditional_weights={
        ("peripheral","new"): {"yes":0.41, "no":0.59},
        ("peripheral","old"): {"yes":0.82, "no":0.18}, # ...
    })
```

```
sim.add_measure_structural("pickup_to_dropoff_min",
    formula="8 + 3.5*distance_km - 4*(algorithm=='new')",
    noise={"sigma":2.0})
```

zone-selection trick: new algo silently drops the periphery

```
sim.inject_pattern("dominance_shift",
    target="algorithm=='new' & zone_tier=='peripheral'",
    col="accepted", params={"drop_rate":0.43})
```

Every behavior the agent must discover is **declared once** in the script.

Math errors are caught at the **SDK level**.

Typed exceptions feed back to the LLM. Three validator layers run after every build.

ATTEMPT 1 · UNBOUNDED SPEEDUP

```
sim.add_measure_structural("pickup_to_dropoff_min",  
    formula="5 - 30*(algorithm=='new')")
```

✗ **Validator L2:** $\min(p \rightarrow d) = -25 < 0 \Rightarrow \text{SDKException}$

↓ **feedback to LLM · ≤ 3 retries**

ATTEMPT 2 · BOUNDED, DISTANCE-AWARE

```
sim.add_measure_structural("pickup_to_dropoff_min",  
    formula="8 + 3.5*distance_km - 4*(algorithm=='new')",  
    noise={"sigma":2.0})
```

✓ **All three layers pass** · 1000 atomic rows written.

THREE-LAYER VALIDATOR

L1

Structural

Schema valid? Types compatible? Every DAG acyclic?

L2

Statistical

Moments in range? Correlations feasible? Bounds respected?

L3

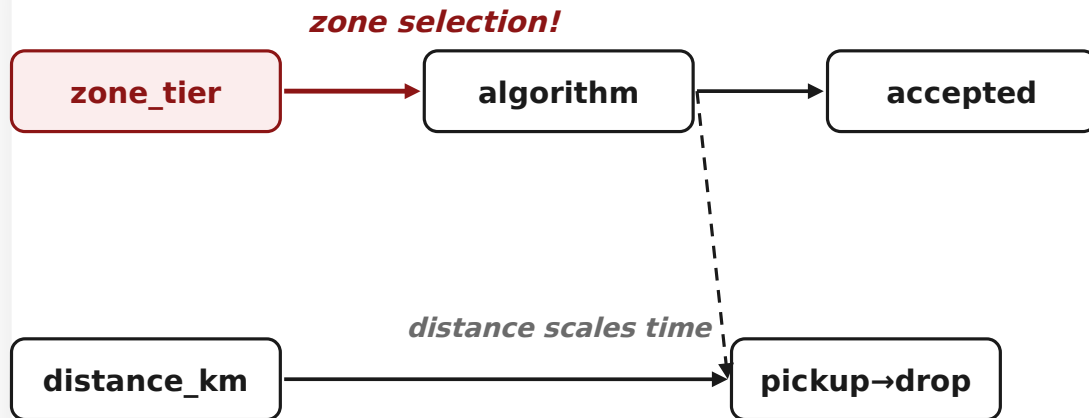
Pattern

Injected trends · drop-outs · outliers all recoverable?

The truth has **structure**.

M is a DAG over typed columns. Atomic rows are samples from it.

SCHEMA DAG · SwiftEats



ATOMIC ROWS · 1 row = 1 order

id	zone	algo	out	p→d
01	ctr	new	✓	16
02	per	new	×	—
03	mid	old	✓	28
04	ctr	new	✓	14
05	per	old	✓	38
06	per	new	×	—
07	mid	new	✓	19

... 1000 rows total

Every aggregation lives **downstream**. Bar totals = pie totals = line totals.

One master table. **Many chart lenses.**

Deterministic SQL projection — 10-30+ coherent tasks per Master Table.

MASTER M · SwiftEats · 1000 atomic rows



LENS A · BAR

Avg time by algorithm

```
SELECT algorithm, AVG(p2d)
FROM M GROUP BY algorithm
```

LENS B · HEATMAP

Accept rate × zone

```
SELECT zone, algo, AVG(accept)
FROM M GROUP BY zone, algo
```

LENS C · FUNNEL

Stage × algorithm

```
SELECT stage, COUNT(*)
FROM M GROUP BY algo, stage
```

Same M. Three lenses. Arithmetic consistency by construction.

A question is a **typed pipeline**.

View → Filter → GroupBy → Sort → ArgMax. Difficulty = number of operators.

Q · *"Which zone tier has the highest delivery time under the new algorithm?"*

M → **Filter**(algorithm='new')
→ **GroupBy**(zone_tier, **AVG**(pickup_to_dropoff_min))
→ **Sort**(desc)
→ **ArgMax**
→ "peripheral" *# 5 ops = Medium difficulty*

16

typed operators

Set · Scalar · Combinator · Bridge

6

chart families · 16 types

Comparison, Trend, Distribution ...

0

LLM calls in Phase 3

deterministic, seed-reproducible

PART · C2

How an agent **investigates** a chart.

Four moves · two costly actions · one bbox-grounded answer.

Read • Reveal • Reconcile • **Debunk.**

Each move is one step of cracking the 18-minute case.

1

READ

the claim, not the number

What two timestamps does the “18 min” span? Which orders are in the denominator?

2

REVEAL

the missing dimension

Stratify by zone: downtown / mid-ring / periphery. Did the new dispatcher abandon distant zones?

3

RECONCILE

conflicting charts coexist

Delivery time fell, but cancellations and refunds rose. What funnel makes both true at once?

4

DEBUNK

the honest yardstick

Re-include every cancel and timeout; widen the window to click→door. Does “18 min” survive?

Five actions. Two are costly.

Budget forces strategy. Every claim must point to a bbox.

ACTION	ROLE	COST	RETURNS
request_view	Pull a new projection	-1 B_v	PNG only
point	Inspect element locally	-1 B_p	bbox feedback
commit_belief	Update case theory	free	confirmed
answer	Submit answer + bbox	free	logged
terminate	Trigger scoring	free	final report

VISUAL-ONLY RETURN

request_view returns PNG, never CSV.

Prevents protocol from degenerating to SQL planning.

BBOX = EVIDENCE RECEIPT

Cross-chart, mandatory.

Grounding is accountability, not decoration.

B1 — READ · the claim, not the number

The agent does not answer 18 min — it asks: between which two timestamps is the 18 min measured, and which orders are in it?

RETURNED VIEW · time-window + denominator filter



570 delivered · in 18 min

430 failed · silently excluded

1 000 orders requested · '18 min' computed only on the accent slice

KEY FINDING

Two narrow choices hide inside '18 min':

pickup → drop only

delivered orders only

SUSPECT BOARD

h₁

Genuine speedup
pending

h₂

Zone selection
pending

h₃

Denominator wash
rising ↑

h₄

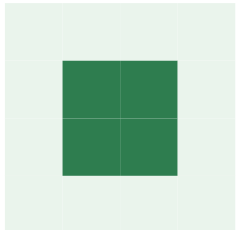
Metric boundary
rising ↑

B2 — REVEAL · the missing zone dimension

B1 showed 430 orders excluded. Now: where on the map did those orders come from?

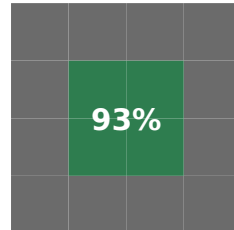
RETURNED VIEW · acceptance × zone × algorithm

Old · city-wide **82%**



OLD · uniform
82%

New · periphery **41%**



NEW · edge
dropped to 41%

KEY FINDING

Downtown is steady; the outer ring collapses.

Old · periphery **82%**

New · periphery **41%**

SUSPECT BOARD

h₁

Genuine speedup
pending

h₂

Zone selection
✓ *confirmed*

h₃

Denominator wash
rising ↑

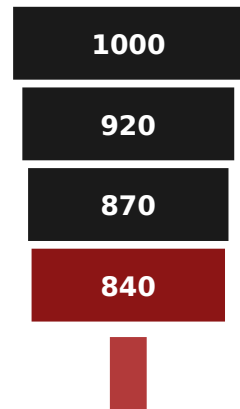
h₄

Metric boundary
rising ↑

B3 — RECONCILE · let both stories coexist

Delivery time fell, refunds rose. Now: what funnel structure lets both be true at the same time?

RETURNED VIEW · requested → assigned → picked up → delivered



160 failed

OLD

success 84%

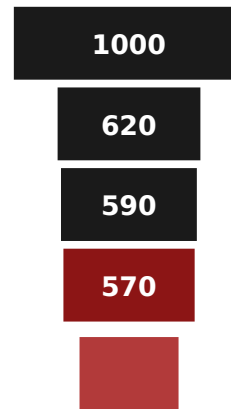
requested

assigned

picked up

delivered

failed



430 failed

NEW

success 57%

KEY FINDING

Both stories coexist on different denominators.

84% → 57%

The 430 failed orders vanish from the headline. Delivered orders really are faster — but they're a different population.

SUSPECT BOARD



Genuine speedup
rising ↑



Zone selection
✓ confirmed



Denominator wash
✓ confirmed



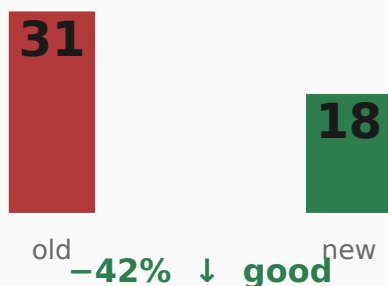
Metric boundary
rising ↑

B4 — **DEBUNK** • the honest yardstick

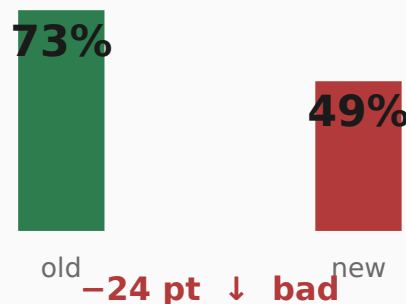
Three beats narrowed the suspects. Now: widen window to click→door and re-include all 1000 — does '18 min' survive?

RETURNED VIEW • headline metric vs. honest yardstick

HEADLINE • pickup → drop (min)



HONEST • click → door on-time (%)



KEY FINDING

On the honest yardstick,
the headline **flips direction**.

73% → 49%

FINAL VERDICT

31 → 18 min = zone selection + denominator wash + metric boundary

SUSPECT BOARD



Genuine speedup
rising ↑



Zone selection
✓ *confirmed*



Denominator wash
✓ *confirmed*



Metric boundary
✓ *confirmed*

Three studies separate **action** from perception.

A

Action Necessity

Model axis $\times$ protocol axis (passive / random / planning).

EXPECTED

Action axis explains more variance than model scale.

B

Evidence Efficiency Pareto

$B_v \in \{2, 4, 8, 16, \infty\}$ on Recovery + Resolution.

EXPECTED

Frontier VLM at $B_v=16$ still below Oracle at $B_v=4$.

C

Channel Ablation

request_view returns PNG only / CSV only / both.

EXPECTED

Decides: chart understanding vs SQL planning.

Punchline: the bottleneck is not perception — it is **exploration strategy.**

From chart reading to **data-world investigation**.

C1

Atomic-grain Code-as-DGP

LLM writes Python over a typed SDK. One Master Table emits 10-30+ tasks via deterministic SQL projection — cross-chart arithmetic by construction.

C2

Active Chart Reasoning protocol

Budgeted visual actions, bbox-as-receipt grounding, four investigation moves: Read · Reveal · Reconcile · Debunk.

C1×C2

LDW-Gym — the infrastructure

One pipeline emits data and protocol together. Private test sets regenerate on demand — structurally immune to contamination.

THANK YOU

Charts are clues.
The task is investigation.

The target — the latent data world.